

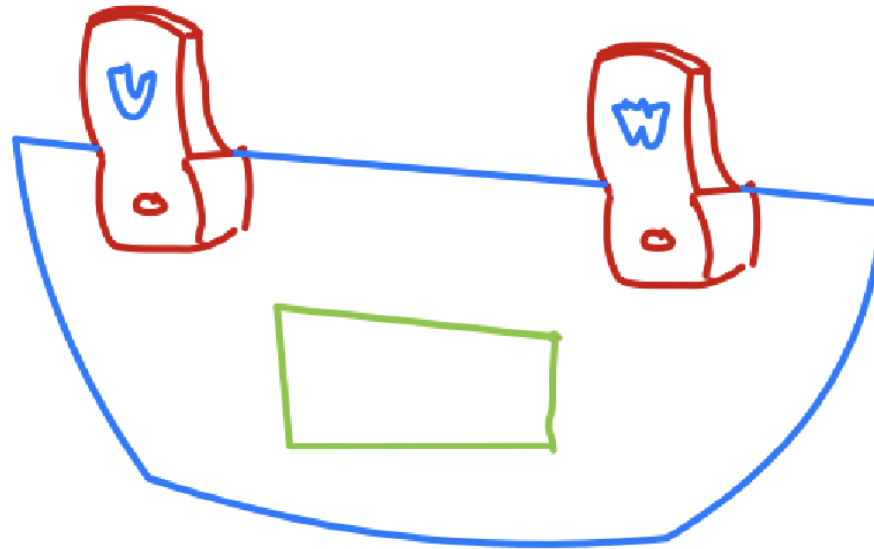
Engineering Methodology

Design Ideation

Concept Sketches

Sketch #1

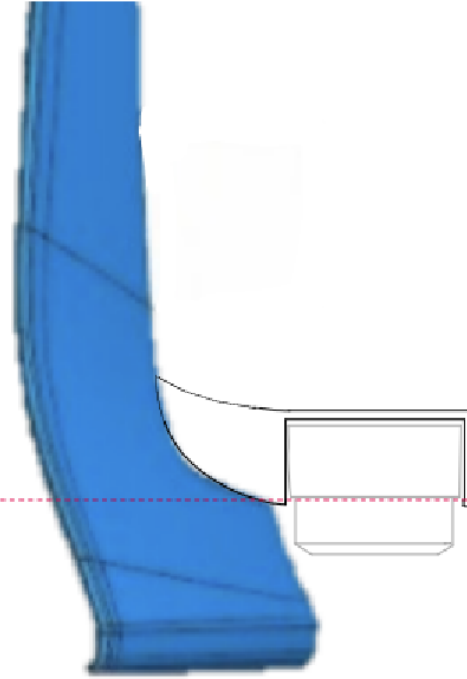
Part



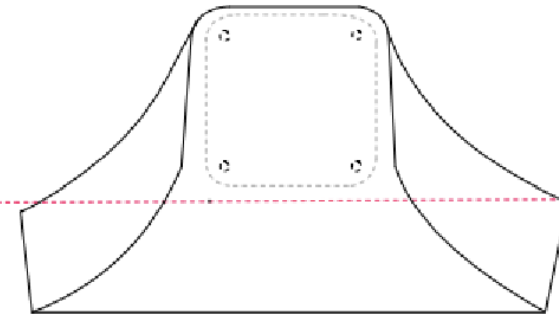
- under the trunk handle
- focus on curvature
- lidar → plate → hook mount
- focus on material selection

Sketch #2

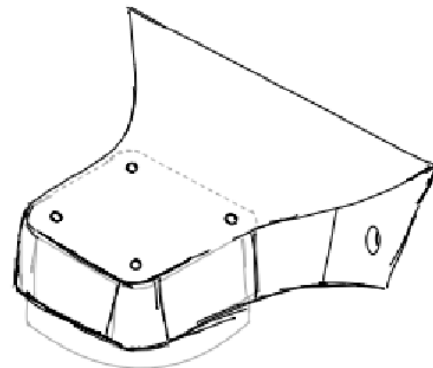
Side View



Top View



Isometric View



Decision Matrix

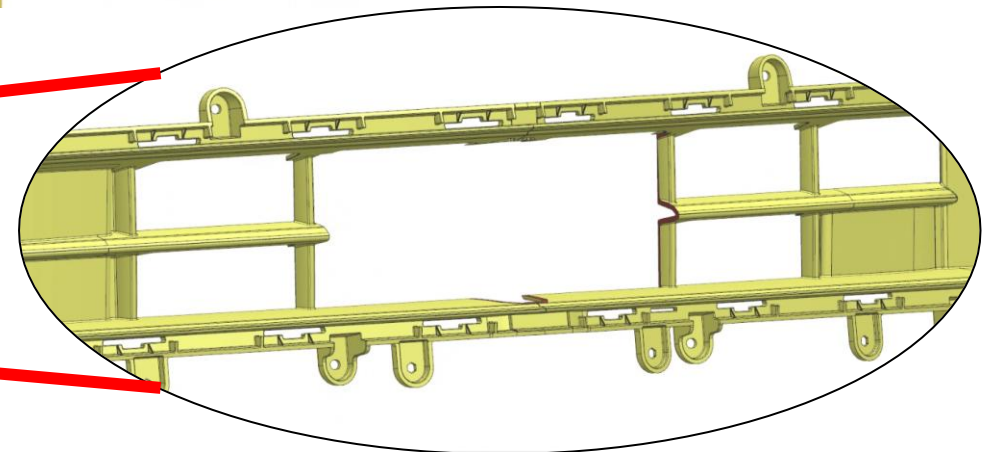
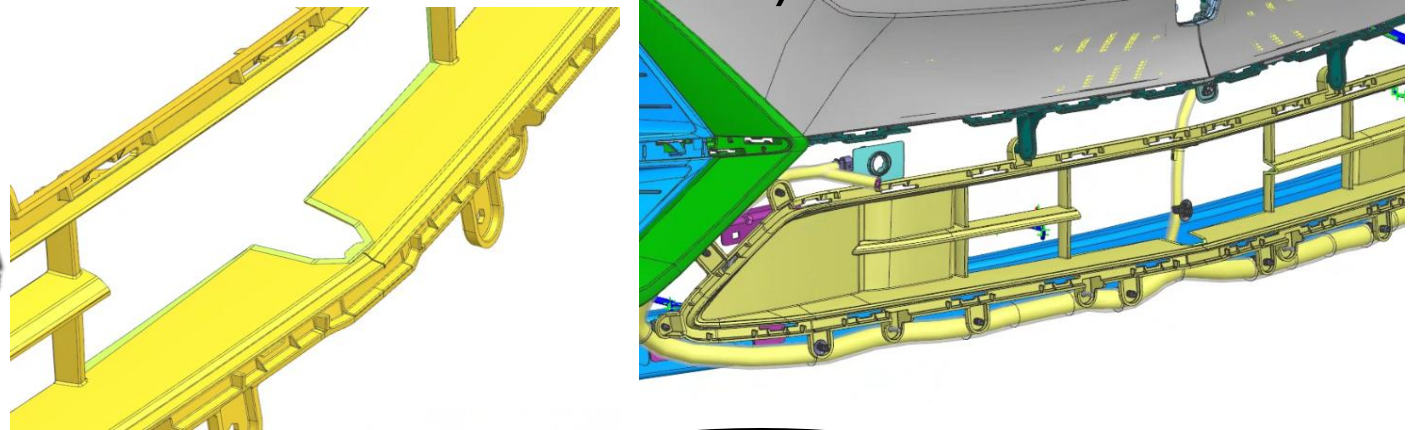
Design Criteria	Sketch #1	Sketch #2	Sketch #3	Sketch #4
Manufacturability	2	3	1	3
Durability	2	3	2	3
Aesthetics and integration with existing vehicle	3	3	1	4
Sensing and FOV	3	4	3	2
Serviceability and ease of access	2	3	2	1
Total Score	12	16	9	13

Design Modifications

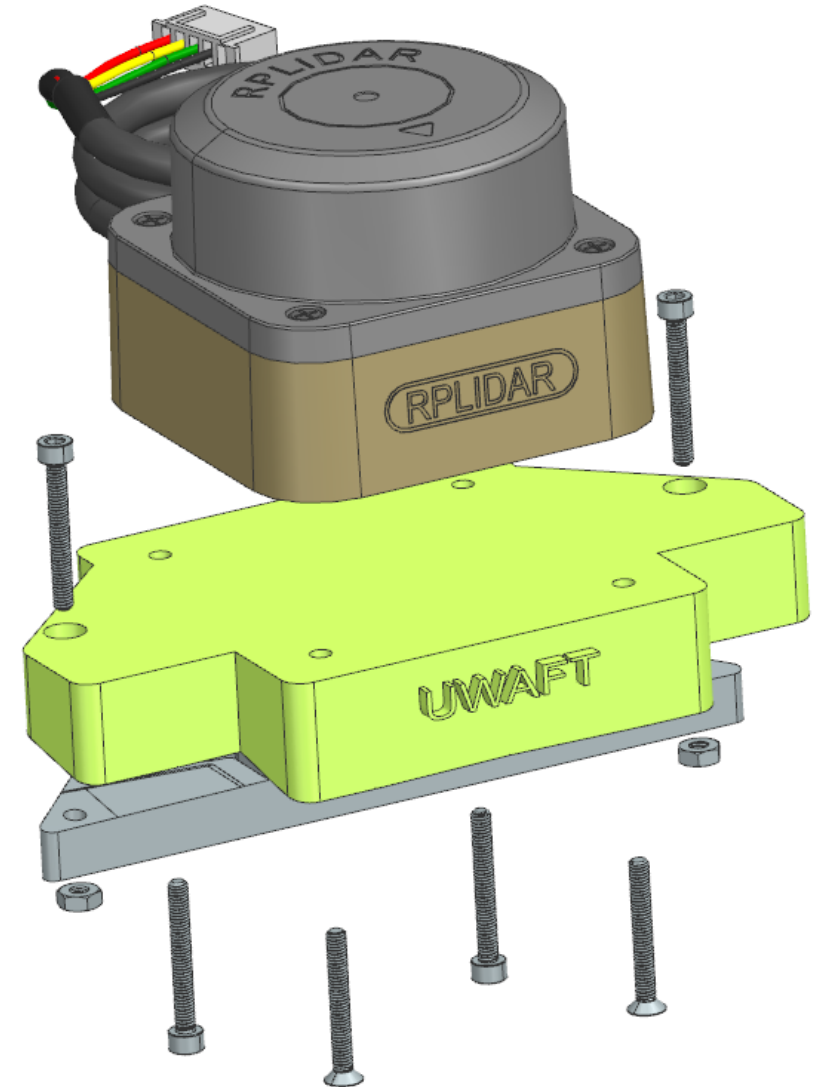
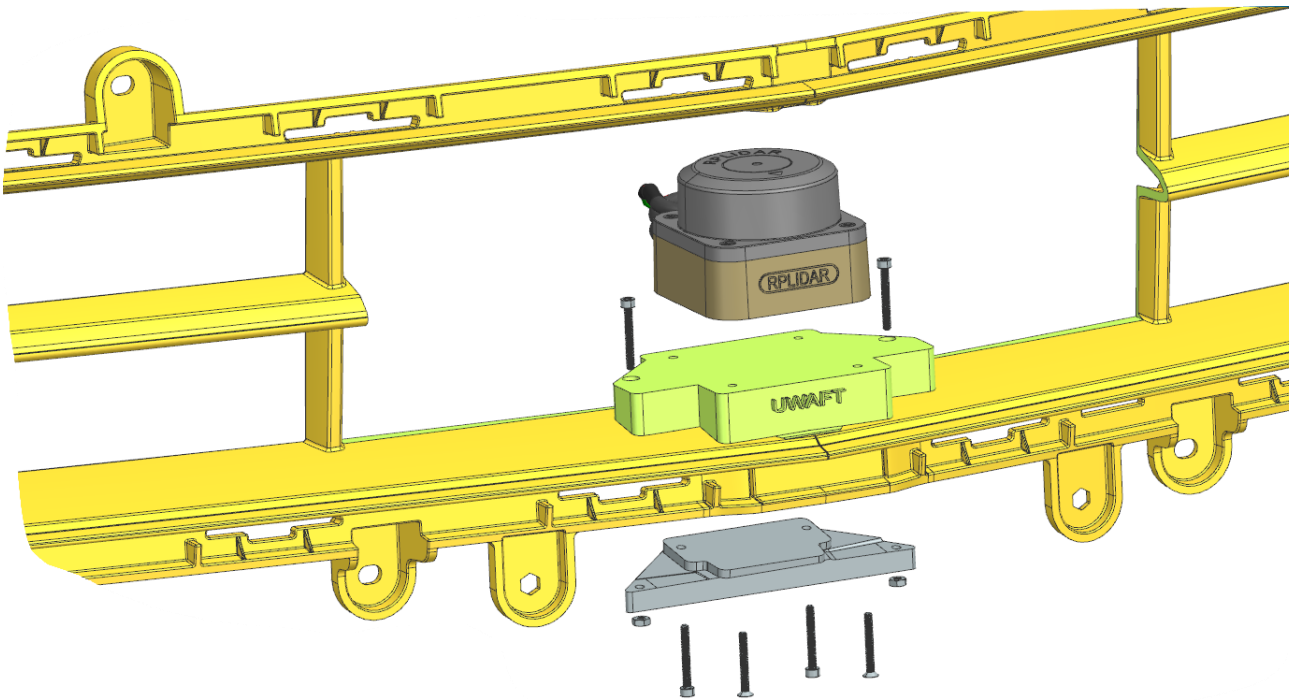
Front & Rear Mount

Front Mount Design

Front Mounting Location: Inside the lower fascia air inlet
(center plastic section is removed for LiDAR clearance)



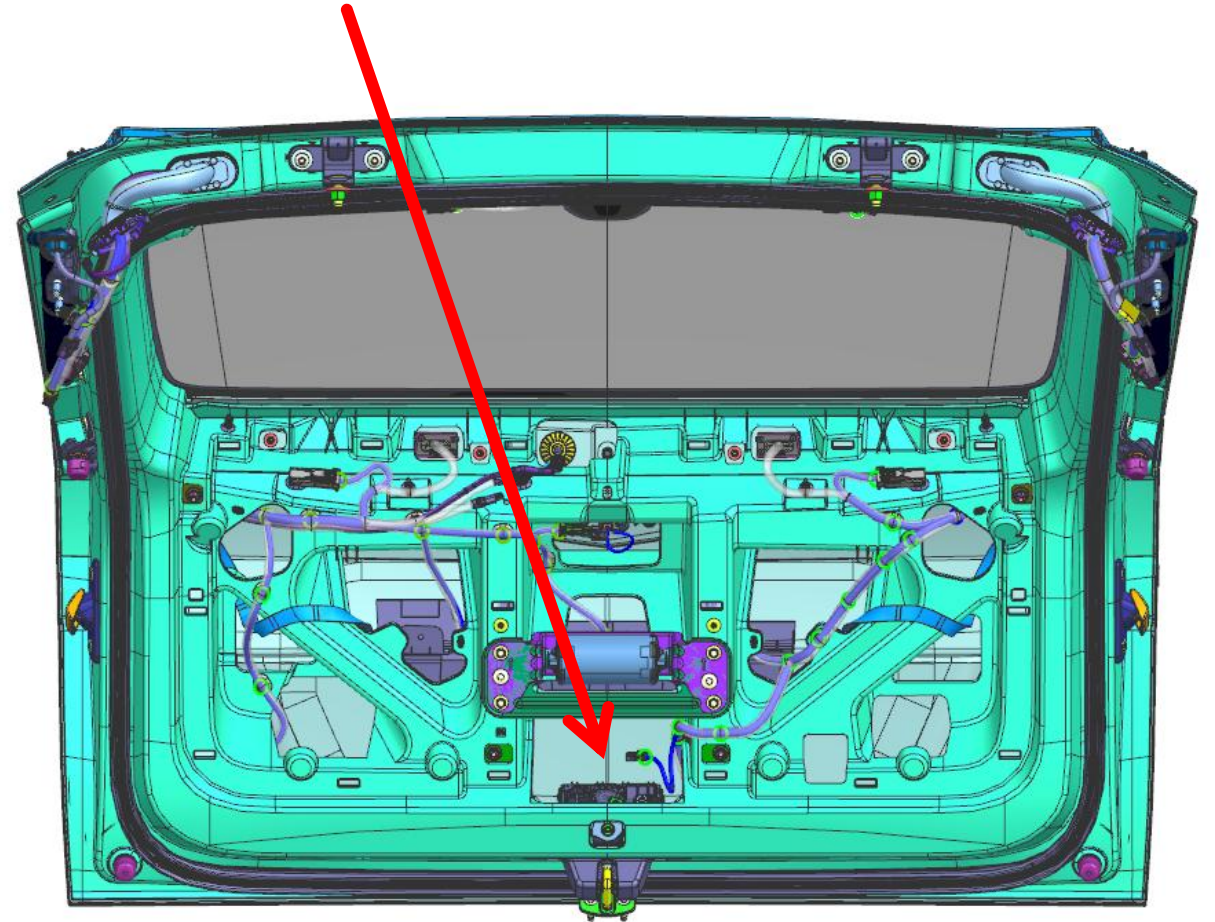
New Front Mount CAD



- LiDAR is mounted parallel to the ground plane
- Mount screws are accessible

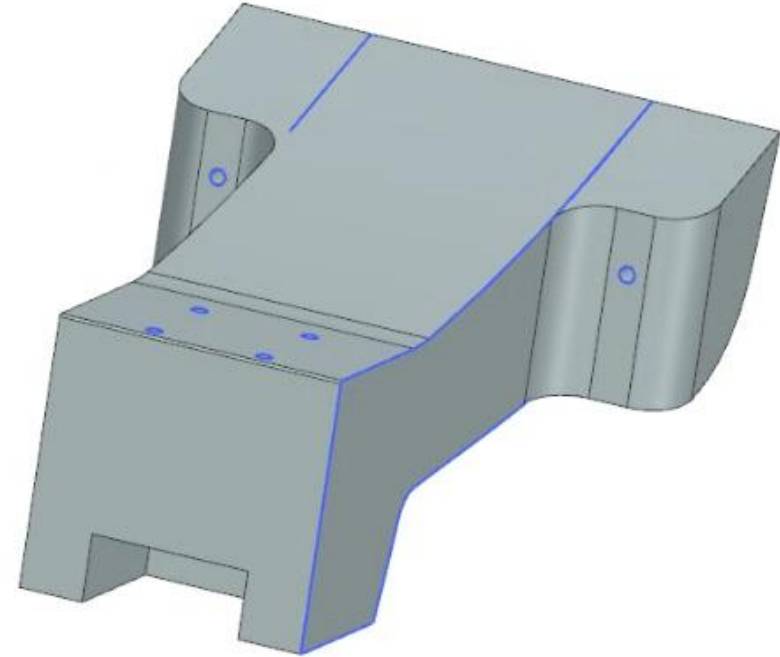
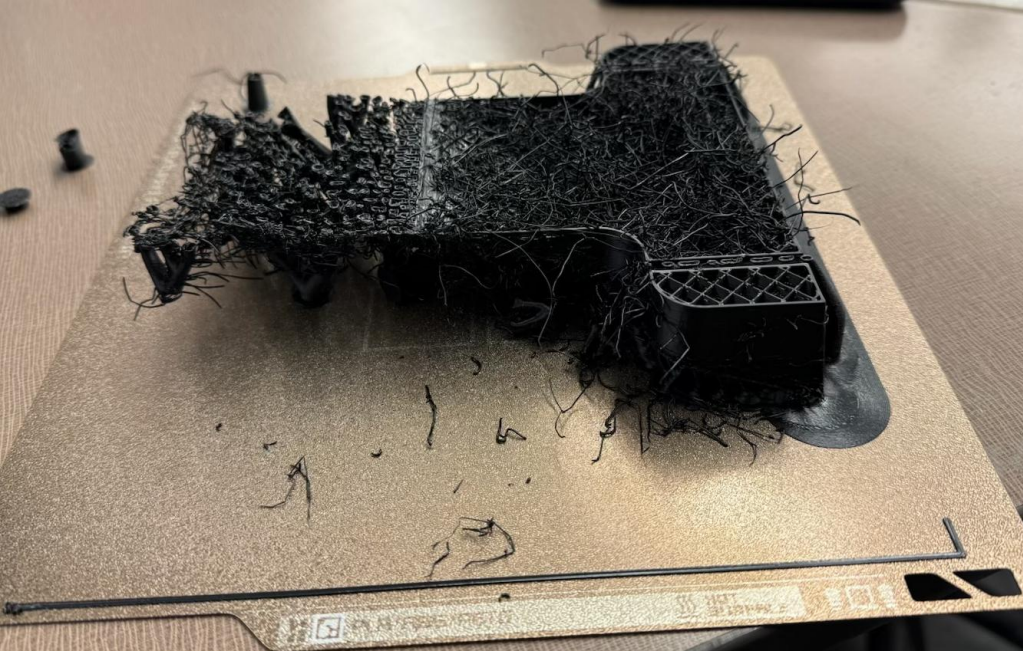
Rear Mount Design

Rear Mounting Design: Underneath the license plate bolting into the trunk



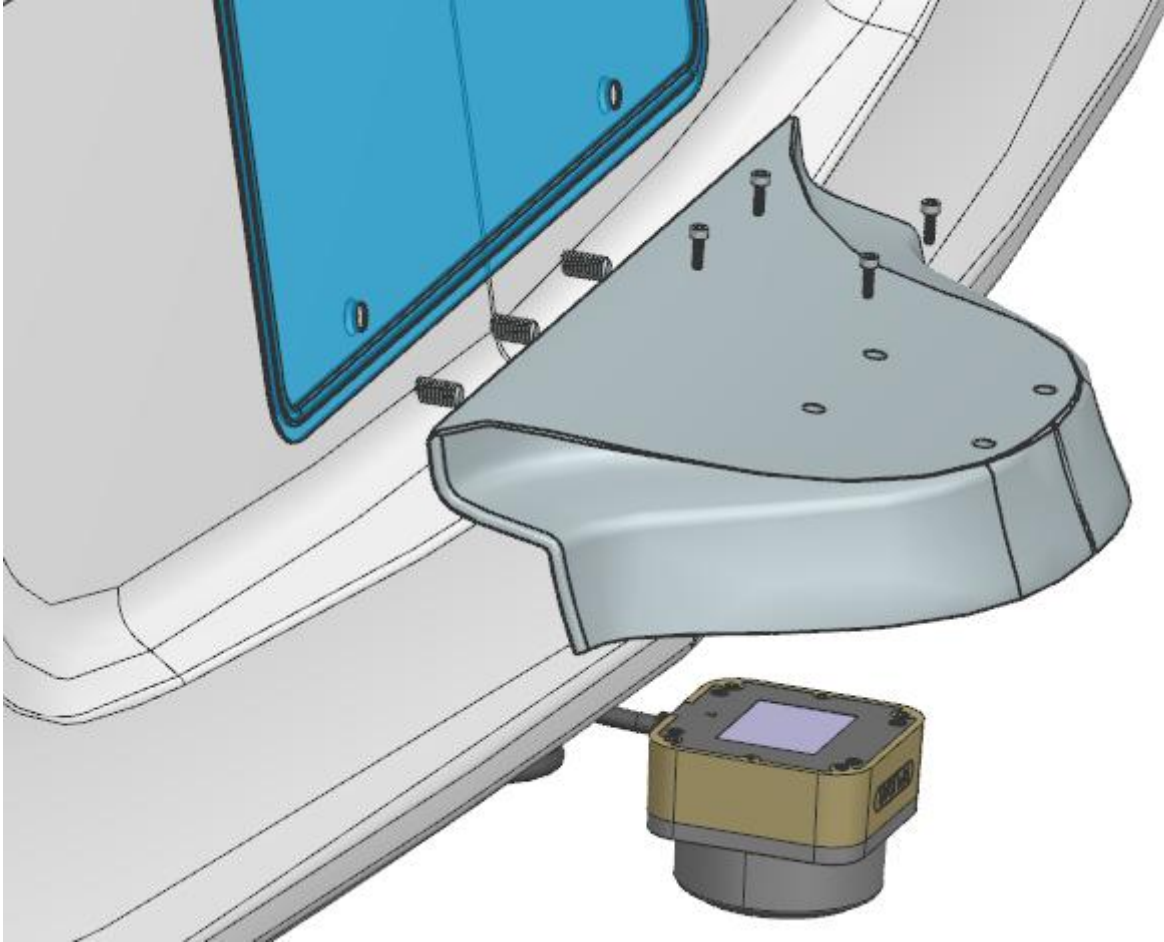
Obstacles & Challenges

1st Revision: Rear Mount Design



- First Iteration of Rear Mount Design failed printing multiple times
- Difficult to print and print time was over 25+ hours

New Rear Mount CAD



ANSYS Simulations

Design Validation

Design Validation

Simulation Parameters

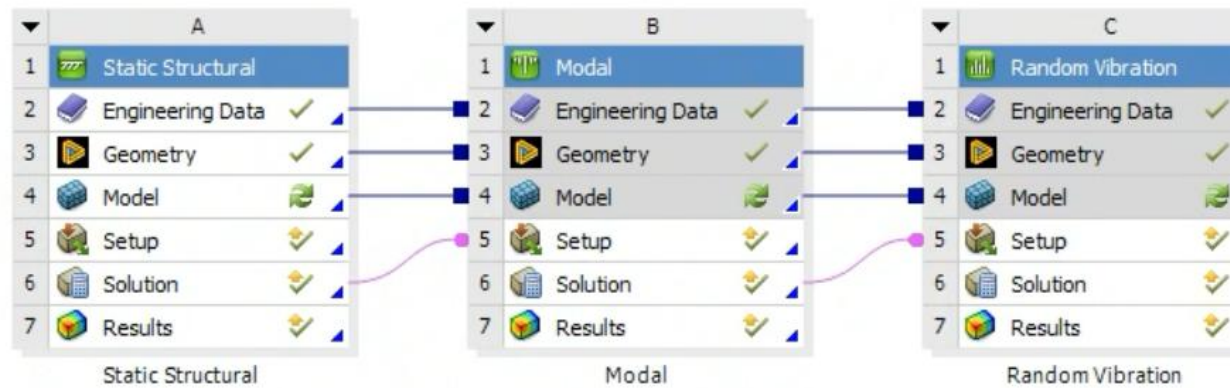


Figure 12: Example for Cross-Axis Acceleration Upper Limit for Random Vibration - Sprung Mass

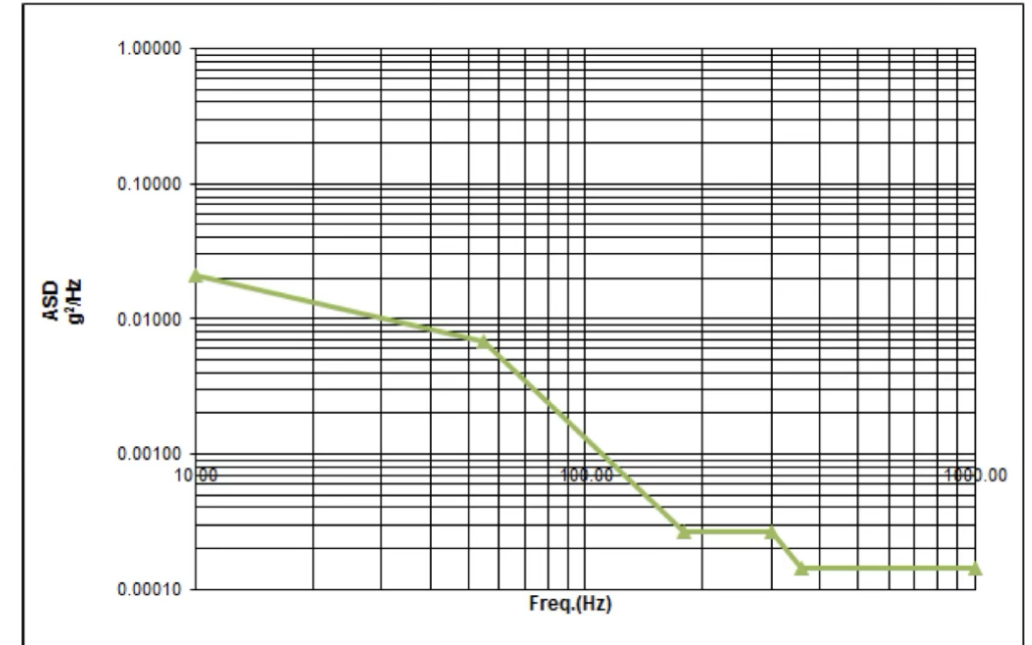


Table 18: Random Vibration Cross-Axis Acceleration Upper Limit for Sprung Mass

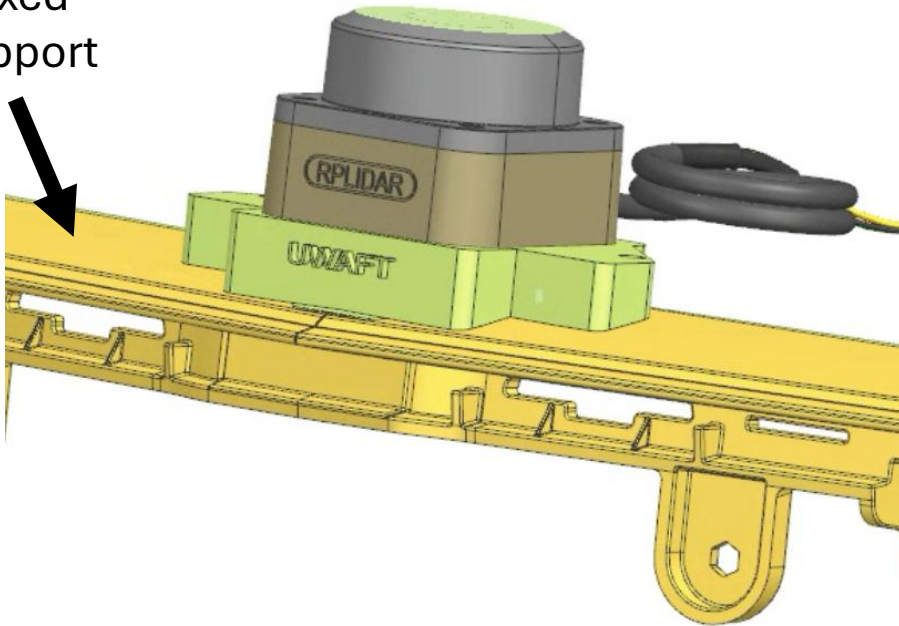
FREQ in Hz	PSD in (m/s²)²/Hz	PSD in g²/Hz
10	2.0098	0.0209
55	0.6543	0.0068
180	0.0253	0.0003
300	0.0253	0.0003
360	0.0136	0.0001
1000	0.0136	0.0001

Effective Acceleration = $8.8 \text{ m/s}^2 = 0.9 \text{ G}_{\text{RMS}}$

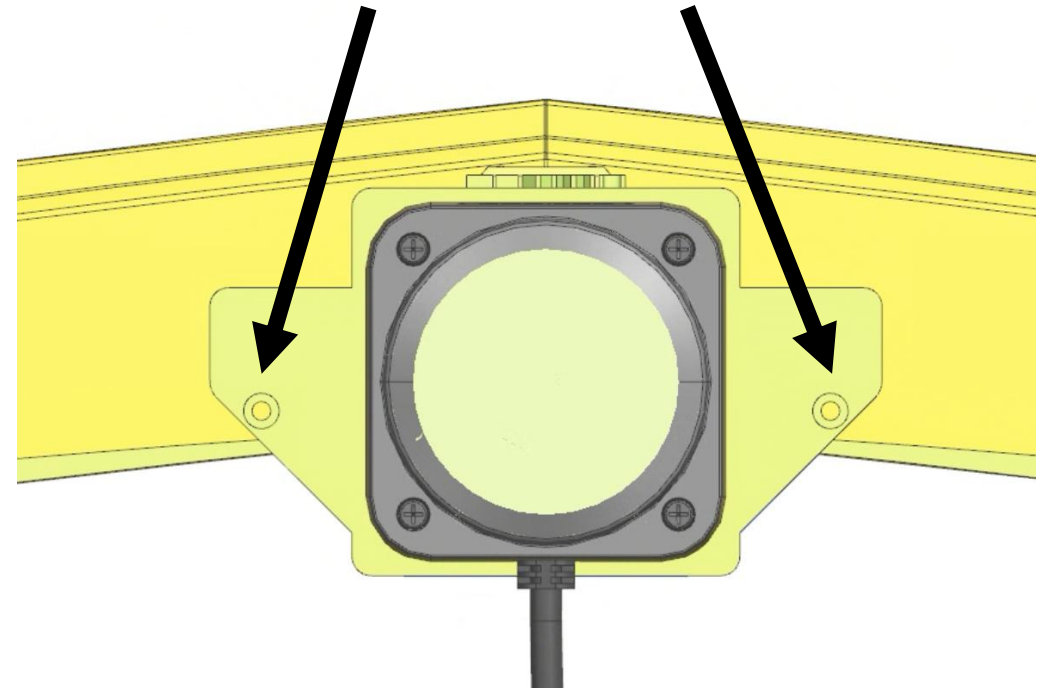
Front Mount Design Validation

Model Setup

Fixed
Support



Bolted Joint
Connection
(Preload Applied)

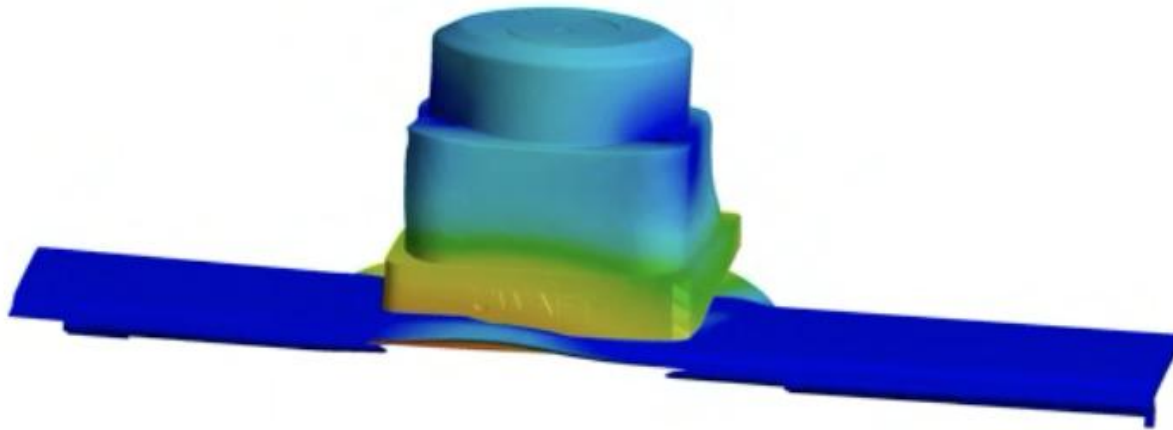
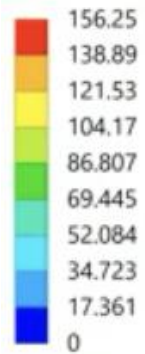


Front Mount Design Validation

Modal Results

Modal

Total Deformation 3
Type: Total Deformation
Frequency: 2051.7 Hz
Unit: mm
Max: 156.25
Min: 0
11/29/2025 9:31 PM



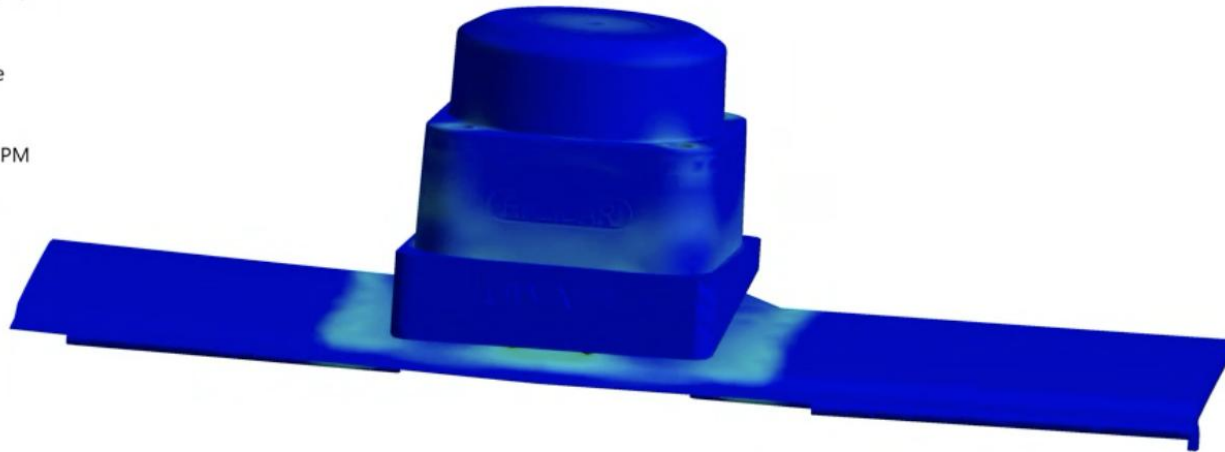
	Mode	☒ Frequency [Hz]
1	1.	1305.4
2	2.	1566.3
3	3.	2051.7
4	4.	2436.8
5	5.	2630.
6	6.	2799.2

Front Mount Design Validation

Random Vibration: Stresses

Random Vibration
Equivalent Stress
Type: Equivalent Stress
Scale Factor Value: 3 Sigma
Probability: 99.73 %
Unit: MPa
Time: 0 s
Custom Obsolete
Max: 0.14827
Min: 6.7373e-8
11/30/2025 8:27 PM

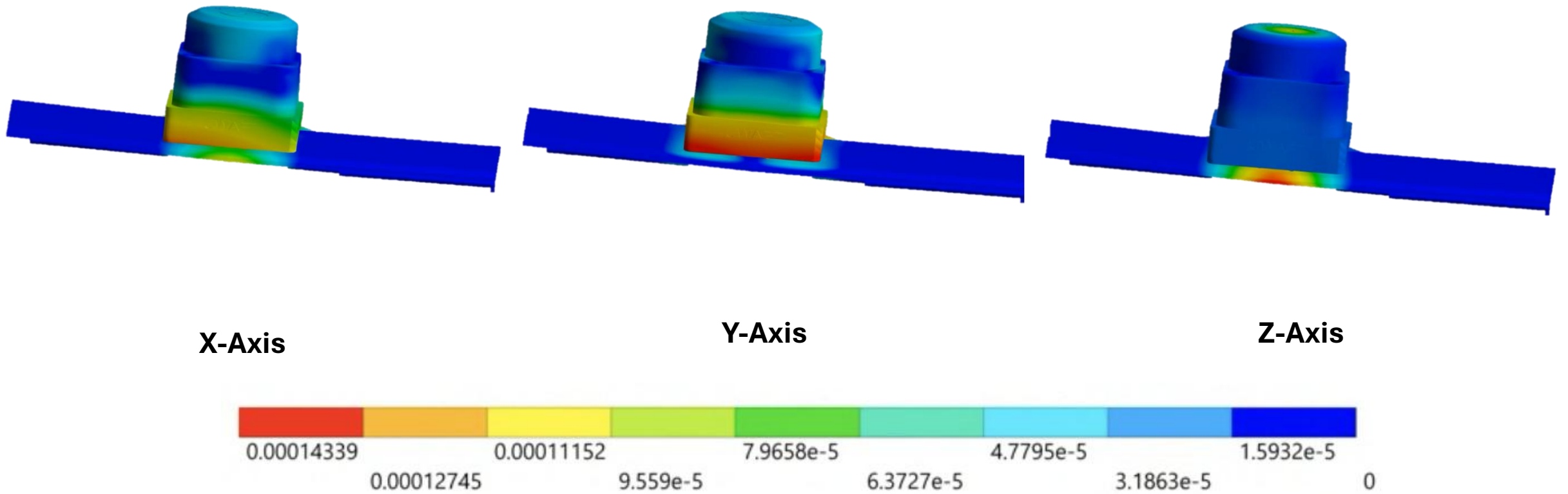
0.065711
0.058415
0.051119
0.043823
0.036528
0.029232
0.021936
0.01464
0.0073442
4.8397e-5



**Bolt mounting
to Bumper**

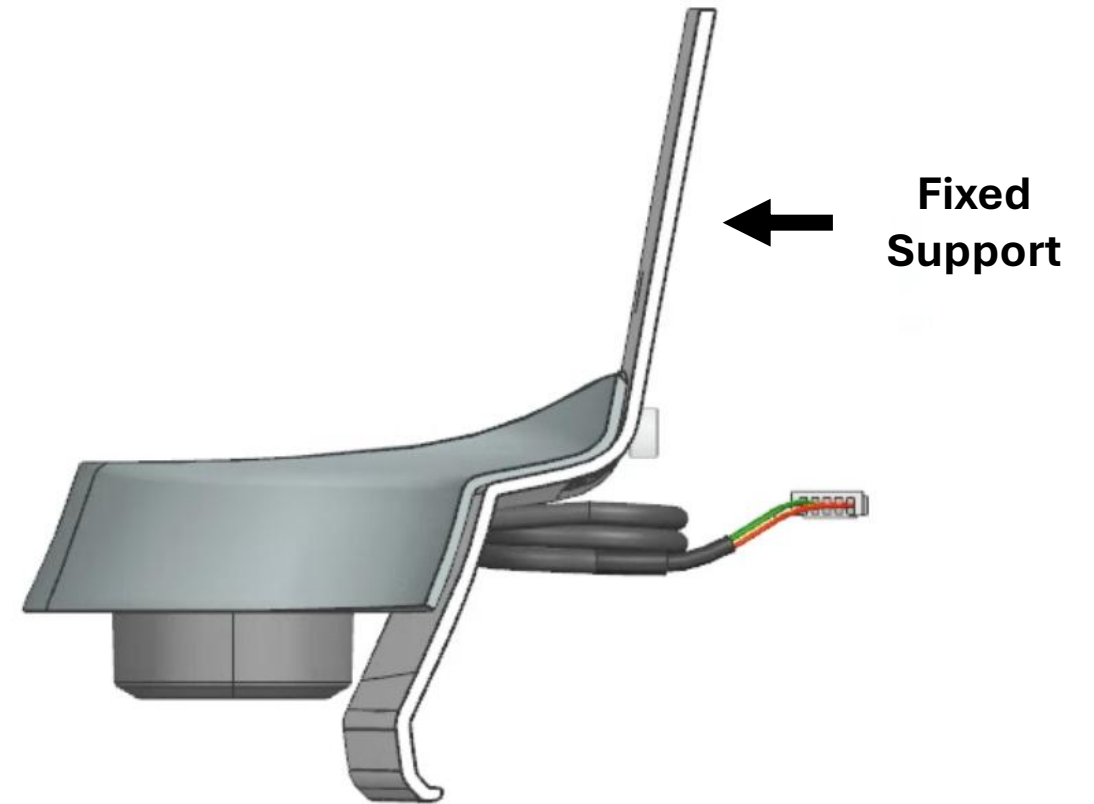
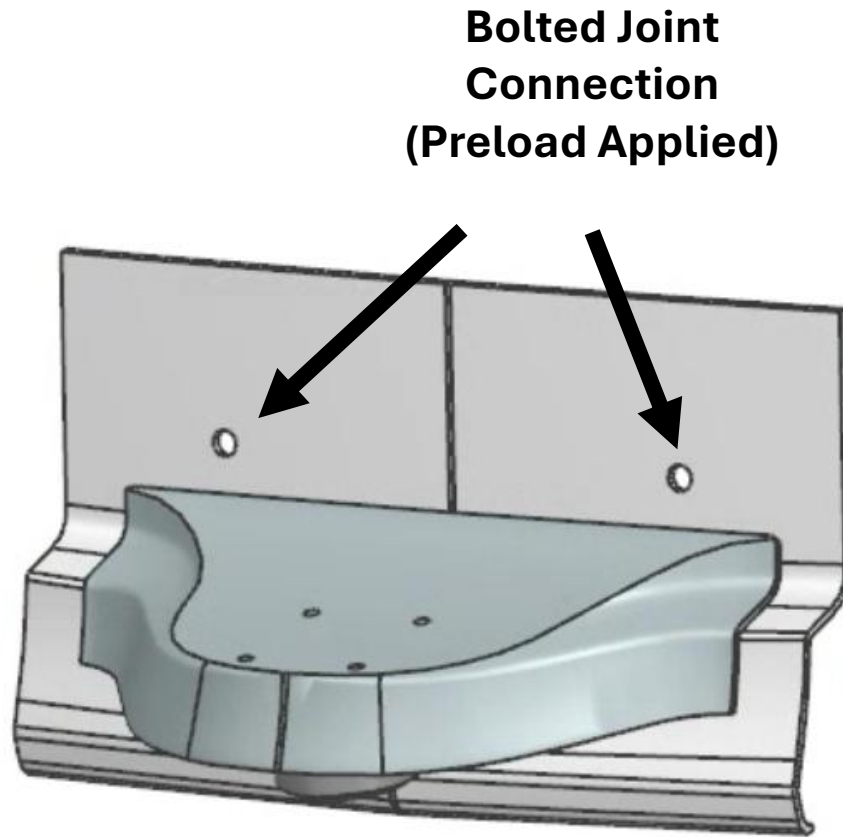
Front Mount Design Validation

Random Vibration: Directional Deformation



Rear Mount Design Validation

Model Setup



Rear Mount Design Validation

Modal Results

B: Modal

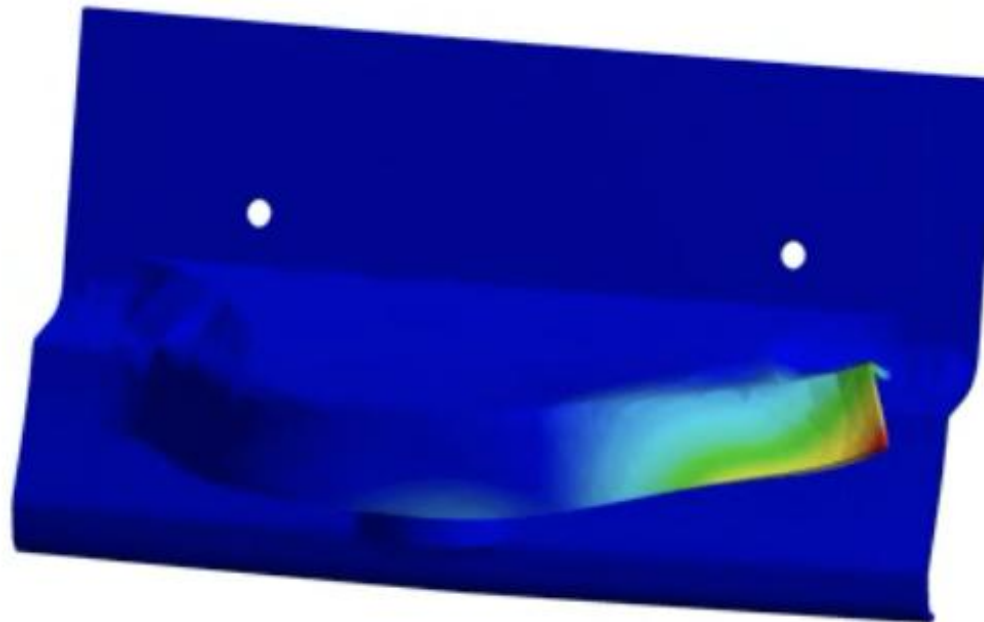
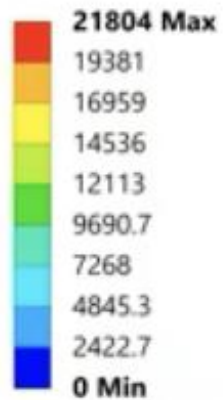
Total Deformation

Type: Total Deformation

Frequency: 575.05 Hz

Unit: mm

2025-11-30 7:56:21 PM

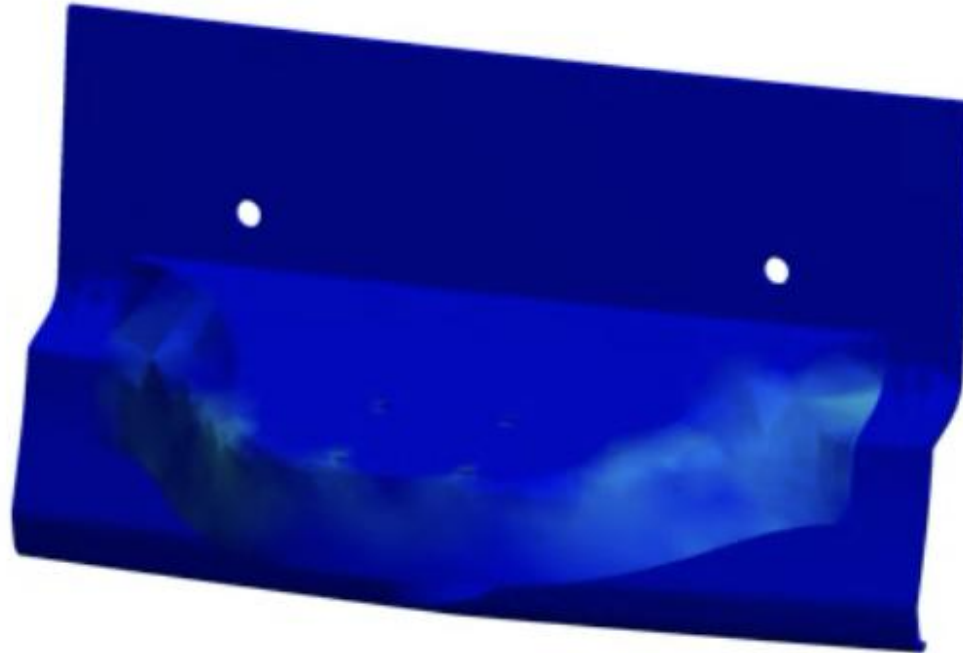
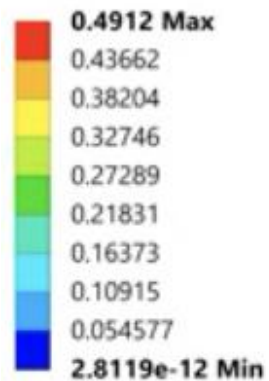


	Mode	☒ Frequency [Hz]
1	1.	575.05
2	2.	656.88
3	3.	805.56
4	4.	959.17
5	5.	997.73
6	6.	1071.9

Rear Mount Design Validation

Random Vibration: Stresses

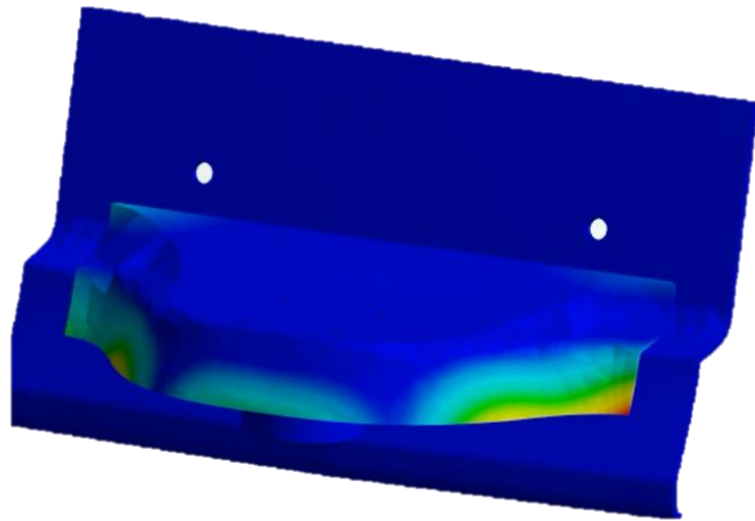
C: Random Vibration
Equivalent Stress
Type: Equivalent Stress
Scale Factor Value: 3 Sigma
Probability: 99.73 %
Unit: MPa
Time: 0 s
2025-11-30 8:00:53 PM



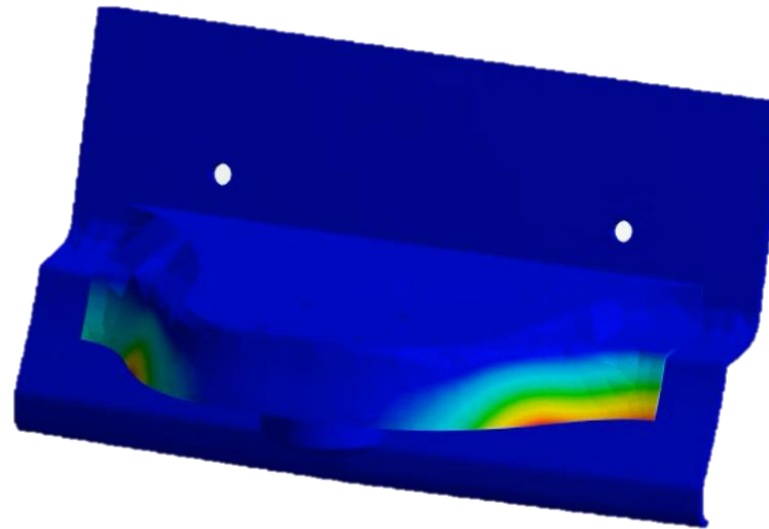
**Bolt mounting
to vehicle**

Rear Mount Design Validation

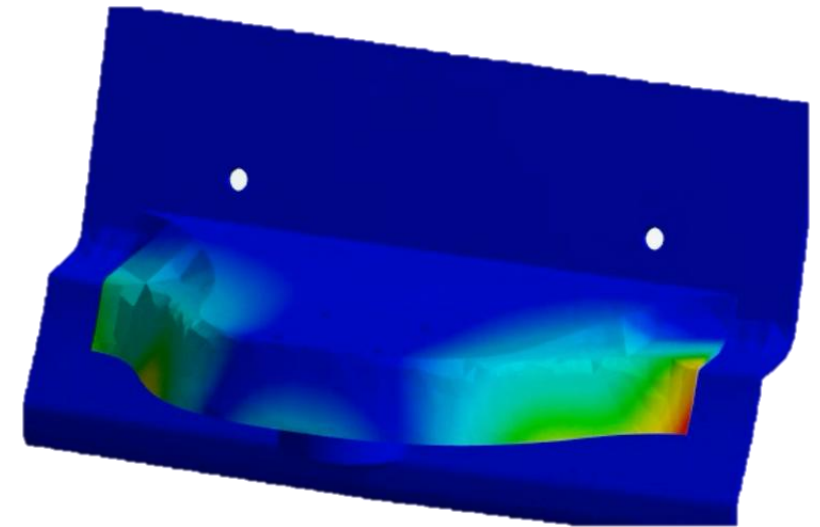
Random Vibration: Directional Deformation



X-Axis



Y-Axis



Z-Axis

